



SALAS O'BRIEN
3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Enclave; Cowart Residential
Location: N/A
Description: Carrington - E, Slab, SunRm, ACH50

Date: November 27, 2024
By: JTH / TJH
House Faces: Worst NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	1,956 ft ²
Volume	17,053 ft ³
Percentage Glass	10 %
Number of Bedrooms	3

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	8581
SEER	14
ACH50	3.0
Mechanical Ventilation	YES
House Outside Air CFM	60

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.056	Attic	2x6	R-18 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.077	Garage Door		R-13	
Roofs	#	U-Value	Description/Notes			
	R1	0.033	Flat Ceiling	2x4	R-30 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed	2x4	R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.34	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			24832 Btu/hr	28313 Btu/hr	
Minimum Nominal Capacity			2.76 Tons	8.30 kW	
Load Density			709 SF/Ton	4.2 W/SF	
Floor Area Served			1956.0 SF	1956.0 SF	
Mechanical Ventilation			60 CFM	60 CFM	
System Size			3.0 Tons	10.0 kW	
Leaving Air Temperature			57.2 deg F	96.3 deg F	
Installed Capacity			652 SF/Ton	5.1 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	41	73	57	
Loft	63	40	51	
Owner Suite	159	216	187	
Owner WIC	18	18	18	
Owner Bath	49	66	58	
Bath	19	28	24	
Bed 2	88	86	87	
Bed 3	88	82	85	
Hall & Closet	18	14	16	
Laundry	47	8	28	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Typical Ceiling Height	9	Cooling Size	1.5	DX Cooling kWh	1683
		Heating Size	1.5	RC Heating kWh	2498
Kneewall Height	1	Best Case	1.5	Total kWh	4182

Cooling Size	1.5	DX Cooling kWh	1627
Heating Size	1.5	RC Heating kWh	2772
Best Case	1.5	Total kWh	4399

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